DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY

University Examinations 2021/2022

SPECIAL/SUPPLEMENTARY EXAMINATION FOR THE DEGREE OF **BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY/ BACHELOR OF SCIENCE IN BUSINESS INFORMATION TECHNOLOGY**

CIT 3102/2206: OBJECT ORIENTED PROGRAMMING II / ADVANCED OBJECT-ORIENTED PROGRAMMING

DATE: **TIME:**

Instructions: Answer Question 1 and Any Other Two

Question 1: (30 Marks)

a) Explain the following terms [5mks]
   i. Abstract class:
   ii. Instance variable
   iii. exception
   iv. byte Code
   v. inheritance

b) Why is java both compiled and interpreted language? [4mks]

c) Describe properties of a constructor method. [3mks]

d) Explain encapsulation and how its achieved in java [4mks]

e) Discuss three different type of errors in java programming [6mks]

f) Compare and contrast
   i. method overloading and method overriding [4mks]
   ii. objects and variables [4mks]

Question Two (20mks)

a) Differentiate between:
   i. layout management and containment hierarchy in java GUI applications [4mks]
   ii. instance variable and a class variable [4mks]

g) explain the differences between datagram communication and streaming communications in socket programming [4mks]

h) Explain two way to achieve persistence of data in a java program [2mks]

b) The code below has two classes.

```
1.  public class Work {
2.   int a=5 , b=0;
3.   public int getratio() {  return this.a/this.b; }
4.  public int div() { int x=0;
5.   try{x= this.a/this.b;}
6.  catch (Exception ex){System.out.printn("can't happen");}
7.   return x;    }
8.   }// end of class work
9.    class WorkTest{
10. public static void main(String args[]){
11.   Work wk=new Work();
12.   try { wk.getRatio();}
13.   catch (Exception ex)
14.   {System.ou.printn("try again");}
15.   try{wk.div;}
16.   catch (Exception e)
17.  {System.out.println("tryingto devide");}
18. }//end of main method
19. }// end of class work test.
```

Required

i. Explain the purpose for the try and catch blocks in lines 5,12 and 15. [2mks]

ii. If the program was run successfully, what would be the out on the screen? [2mks]

iii. If the program was run successfully, which of the three catch blocks line 6,line 13 and line 16 will not be executed and why? [2mks]

Question three (20mks)

a) Discuss any two properties of top level containers in java language [2mks]

b) Explain any three top level containers and their uses [6mks]

c) Consider the code below and answer the questions that follow.

```
1  import javax.swing.*;
2  public class PushMe extends JFrame{
3  public PushMe(){
4  JFrame myJFrame = new JFrame("My Swing window");
```

```
5  JButton button = new JButton("Push me!");
6  myJFrame.getContentPane().add(button);
7  myJFrame.pack();
8  myJFrame.setVisible(true);
9  }// end of constructor
10 public static void main (String[] args) {
11 // create a single PushMe object
12 PushMe myPushMe = new PushMe();
13 }; // end of main
14 };// end of PushMe
```

Required:

i) Sketch the likely output for the program. [3mks]

ii) Edit the code above to include the following:

a. Add a label with text “start” displayed on it, next to the button. [2mks]

b. Add functionality such that, when the button is clicked, the label displays the message “clicked”. [7mks]

Question Four (20mks.)

a) Assume that you are a manager of an auto garage that deals with car maintenance and repair. Among the records you keep in your books are the details of your clients car including: **regno, color, date serviced and amount of money paid**. You are to create a java program that stores this details into a computer.

i) Write a class Car with the following properties:

1. All the attributes to include regno, color, date serviced and amount of money paid. [2mks]
2. two constructor methods; one empty [3mks]
3. a method for saving the car details into a file “cars.dat” [5mks]
4. a method that reads the details from the file “cars.dat” and reassigns them to the corresponding attributes [5mks]
5. add a method that would save the car details to a database called “mygaragedb” and into a table called “CAR”. [5mks]

END.